

BUDETEN® INHALER 200mcg*Budesonide**pressurised metered-dose inhaler/suspension***Composition***Each metered dose inhaler contains budesonide BP 200mcg.***Description**

Budeten® Inhaler 200 mcg is a pressurised metered dose inhaler containing not less than 200 actuations of white or almost white suspension of *budesonide* filled in a 19 ml aluminium canister fitted with a 50 microlitres plastic and elastomeric metering valve with a white polypropylene actuator and light pink polypropylene dust cap and packed into individual printed carton. Each actuation delivers 200 mcg of *budesonide*.

Pharmacodynamics

Budesonide is a glucocorticoid with high topical anti-inflammatory effects. *Budesonide's* activity, measured as its affinity for glucocorticoid receptors is approximately 15 times higher than that of prednisolone.

The exact mechanism of action of glucocorticoid in the treatment of asthma is not fully understood. Anti-inflammatory effects such as inhibited release of inflammatory mediators and inhibition of cytokine-mediated immune response are probably important.

Clinically, *budesonide* reduced bronchial obstruction during both the early and the late phase of an allergic reaction. In hyper-reactive patients, *budesonide* reduces the histamine and metacholine reactivity in the airways.

Pharmacokinetics**Absorption**

Peak plasma concentration (4.0 nmol/L after a dose of 800g) of *budesonide* occurs within 30 minutes following oral inhalation of *budesonide*. Maximum plasma concentration and area under the plasma concentration time profile increase linearly with dose but are slightly (20-30%) higher following repeated doses (3 weeks treatment) than after a single dose.

Distribution

Volume of distribution of inhaled *budesonide* is approximately 3L/kg. Plasma protein binding averages 85-90%.

Biotransformation

Budesonide undergoes an extensive degree (~90%) of biotransformation on first passage through the liver to metabolites of low glucocorticosteroid activity. The glucocorticosteroid activity of the major metabolites, *6β-hydroxybudesonide* and *16α-hydroxyprednisolone*, is less than 1% of that of

budesonide. The metabolism of *budesonide* is primarily mediated by CYP3A, a subfamily of cytochrome P450.

Excretion

The metabolites of *budesonide* are excreted as such or in conjugated form mainly via the kidneys. No unchanged *budesonide* has been detected in the urine. *Budesonide* has high systemic clearance (approximately 1.2 L/min) in healthy adults, and the terminal half-life of *budesonide* after IV dosing averages 2-3 hours. The pharmacokinetics of *budesonide* is proportional to the dose at relevant dosages.

Linearity/non-linearity

The kinetics of *budesonide* is dose-proportional at clinically relevant doses.

Paediatric safety data

Budesonide has a systemic clearance of approximately 0.5 L/min in 4-6 years old asthmatic children. Per kg body weight children have a clearance which is approximately 50% greater than in adults. The terminal half-life of *budesonide* after inhalation is approximately 2.3 hours in asthmatic children. This is about same as in healthy adults. The pharmacokinetics of *budesonide* in children and in patients with impaired renal function are not known. Exposure to *budesonide* may be greater in patients with hepatic disease.

Indication

Bronchial asthma.

Recommended Dosage

For oral inhalation. The dosage of *Budeten® Inhaler* is individual.

Initially, at the beginning of inhaled corticosteroid therapy, for therapy during periods of severe asthma or when scaling down or withdrawing oral corticosteroids the dosage should be:

Children aged 2-7 years:
200-400 micrograms per day divided into 2 administrations.

Children aged 7 years and older:
200-800 micrograms per day divided into 2 administrations.

Adults:

200-1600 micrograms per day divided into 2 administrations.

For maintenance treatment administration twice daily, morning and evening, is usually sufficient. The maintenance dose should be the lowest possible. Following a single dose an effect may be expected

after a few hours. The full therapeutic effect is only achieved after a few weeks of treatment.

Mode of Administration

Oral Inhalation only. To minimise the risk of oropharyngeal candida infection, the patient should rinse their mouth with water after inhaling.

Contraindications

Hypersensitivity to *budesonide*.

Warnings and Precautions

Patients should be advised on the need to take the medication regularly as advised by their doctors, and that *Budeten® Inhaler* is not designed to relieve acute asthmatic symptoms, of which an inhaled short-acting bronchodilator is needed.

Special care must be given to patients who are transferred from systemic to inhaled corticosteroid, since they may remain at risk of impaired adrenal function. Patients who have required high dose emergency corticosteroids may also be at risk and these patients, when exposed to severe stress may exhibit signs and symptoms of adrenal insufficiency. Additional systemic corticosteroid treatment should be considered during periods of stress or elective surgery.

Systemic effects of inhaled corticosteroids may occur, particularly at high doses prescribed for prolonged periods. These effects of inhaled corticosteroid are less likely to occur than with oral corticosteroids.

Possible systemic effects include Cushing's syndrome, Cushingoid features, adrenal suppression, growth retardation in children and adolescents, decrease in bone mineral density, cataract, glaucoma, and more rarely, a range of psychological or behavioural effects including psychomotor hyperactivity, sleep disorders, anxiety, depression or aggression (particularly in children). It is important to titrate the dose of inhaled corticosteroids to the lowest dose at which effective control of asthma is maintained.

Reduced liver function affects the elimination of corticosteroids, causing lower elimination and higher systemic exposure. Be aware of possible systemic side effects.

Concomitant use of *ketoconazole*, *itraconazole*, HIV protease inhibitor or other CYP3A inhibitors including *cobicistat*-containing products should be avoided.

Special care is necessary in patients with active or quiescent pulmonary tuberculosis, and in patients with fungal or viral infections in the airways.

Oral candidiasis may occur during therapy with inhaled corticosteroids. The infections may require treatment with appropriate antifungal treatment and in some patients, discontinuation of treatment may be necessary.

As with other inhalation therapy, paradoxical bronchospasm may occur with an immediate increase wheezing after dosing. Inhaled *budesonide* should be discontinued immediately the patient assessed and alternative therapy instituted if needed. Contact a physician for other important therapy if treatment with *Budeten® Inhaler* has diminished, as repeated inhalations for severe asthma attacks must not delay the initiation of other important therapy. A course of oral steroid for a short period should be supplemented in the event of an acute deterioration of treatment.

During the transfer from oral steroids to inhaled *budesonide*, patients may experience the return of previous symptoms such as muscle and joint pain. In these cases, temporary increase of oral steroid dose may sometimes necessary. If, in isolated cases, fatigue, headache, nausea, vomiting or similar symptoms occur, a generally unsatisfactory effects of the steroid should be suspected. When systemic steroid treatment is replaced by inhaled steroid, allergies such as rhinitis and eczema are sometimes revealed, which were previously controlled by the systemic treatment.

Influence on growth

It is recommended that the height of children receiving prolonged treatment with inhaled corticosteroids is regularly monitored. If growth is slowed, therapy should be re-evaluated with the aim of reducing the dose of inhaled corticosteroid, if possible, to the lowest dose at which effective control of asthma is maintained. The benefit of the corticosteroid therapy and the possible risk of growth suppression must be carefully weighed. In addition, consideration should be given to referring the patient to a paediatric respiratory specialist.

Pneumonia in patients with COPD

An increase in the incidence of pneumonia, including pneumonia requiring hospitalisation, has been observed in patients with COPD receiving inhaled corticosteroids. There is some evidence of an increased risk of pneumonia with increasing steroid dose but this has not been demonstrated conclusively across all studies. There is no conclusive clinical evidence for intra-class differences in the magnitude of the pneumonia risk among inhaled corticosteroid products. Physicians should remain vigilant for the possible development of pneumonia in patient with COPD as the clinical features of such infections overlap with the symptoms of COPD exacerbations. Risk factors for pneumonia in patients with COPD include current smoking status, older age, low body mass index (BMI) and severe COPD.

Visual disturbances

Visual disturbances may be reported with systemic and topical corticosteroid use. If a patient presents with blurred vision or other visual disturbances, the patient should be considered for referral to an ophthalmologist for evaluation of possible causes

which may include cataract, glaucoma or rare disease such as central serous chorioretinopathy (CSCR) which has been reported after use of systemic and topical corticosteroids.

Effect on ability to drive and use machines.

Budesonide has no influence on the ability to drive or use machines

Interactions with Other Medicaments

The metabolism of *budesonide* is primarily mediated by CYP3A, inhibitors of this enzyme, e.g. *ketoconazole*, *itraconazole*, HIV protease inhibitors and other CYP3A inhibitors, including *cobicistat*-containing products, is expected to increase the risk of systemic side-effects. The combination should be avoided unless the benefit outweighs the increased risk of systemic corticosteroid side-effects, in which case patients should be monitored for systemic corticosteroid side-effects. If the combination is unavoidable, the period between treatments should be long as possible and a reduction of *budesonide* dose could also be considered.

Limited data about this interaction for high-dose inhaled *budesonide* indicated that marked increases in plasma levels (on average four-fold) may occur if *itraconazole* 200mg once daily is administered concomitantly with inhaled *budesonide* (single dose of 1,000mcg).

Raised plasma concentrations of and increased effects of corticosteroids have been observed in women treated with oestrogens and contraceptive steroids, but not with low dose combination oral contraceptives.

Because of adrenal function may be suppressed, an ACTH stimulation test for diagnosing pituitary insufficiency might show false results (low values).

Fertility, Pregnancy and Lactation

Pregnancy

Treatment with *budesonide* did not show any increased risk of malformations. During pregnancy the aim must be the lowest effective dose of *budesonide* and the risk of a worsened asthmatic state must be taken into consideration.

Lactation

Budesonide is excreted in breast milk. At therapeutic dose of *budesonide* inhalation, no effects on suckling child are anticipated and can be used during breast-feeding.

Side Effects

The following definitions apply to the incidence of undesirable effects: Frequencies are defined as: very common ($\geq 1/10$); common ($\geq 1/100$ to $< 1/10$); uncommon ($\geq 1/1,000$ to $< 1/100$); rare ($\geq 1/10,000$ to $< 1/1,000$); very rare ($< 1/10,000$).

Adverse Drug Reactions (ADR) by System Organ Class (SOC) and Frequency

SOC	Frequency	Adverse Drug Reaction
Infections and infestations	Common	Candida infections in the oral cavity and throat. Pneumonia (in COPD patients)
Immune system disorders	Rare	Immediate and delayed hypersensitivity reactions including rash, contact dermatitis, urticaria, angioedema and anaphylactic reaction
Endocrine disorders	Rare	Signs and symptoms of systemic corticosteroid effects, including adrenal suppression and growth retardation*
Eye disorder	Uncommon	Vision blurred
	Unknown	Glaucoma Cataract
Psychiatric disorders	Rare	Restlessness Nervousness Depression Behavioural changes (predominantly in children)
	Unknown	Sleep disorders Anxiety Psychomotor hyperactivity Aggression
Respiratory, thoracic and mediastinal disorders	Common	Cough Throat irritation
	Rare	Bronchospasm Dysphonia Hoarseness
Skin and subcutaneous tissue disorders	Rare	Bruising

* Growth should be monitored regularly in paediatric patients.

On account of the risk of candida infections in the oral cavity and throat the patient must rinse the mouth with water after every dose.

Overdose and Treatment

Acute overdosage with *Budeten® Inhaler* even in excessive doses, is not expected to be a clinical

problem. However, if used chronically in high doses, hypercortisolism and adrenal suppression may occur.

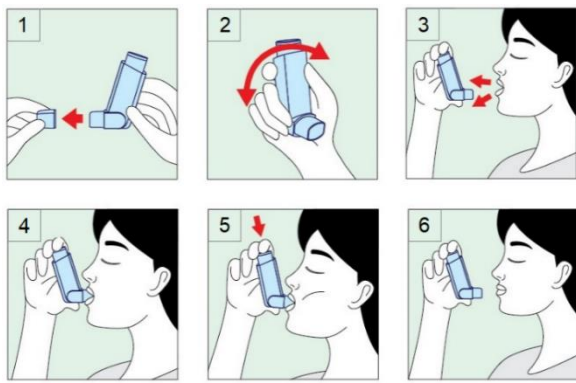
No special emergency action needs to be taken. Treatment with *Budeten® Inhaler* should be continued at the recommended dose to control the asthma.

Instructions How To Use Your Inhaler

Preparing your inhaler

Before using your new *Budeten® Inhaler* for the first time, or if you have not used your inhaler for a minimum of 2 days, remove the dust cap from the actuator and release one puff into the air.

Using your inhaler



1. Remove the dust cap by gently pressing and pulling it away from the actuator and check to ensure that the mouthpiece is clean.
2. Shake the inhaler to ensure that the contents of the inhaler are evenly mixed.
3. Hold the inhaler upright between index finger and thumb, with your thumb on the base of the mouthpiece.
4. Breathe out as far as comfortable and then place the mouthpiece in your mouth between your teeth and close your lips around it. Do not bite it.
5. Tilt your head slightly backwards. Start to breathe in slowly through your mouth. Just after starting to breathe in, press down the canister using your index finger to release the medication while still breathing in steadily and deeply.
6. While holding your breath, remove the inhaler from your mouth. Continue holding your breath for 10 seconds. Then, breathe out through your mouth slowly.
7. If you are to take another puff, keep the inhaler upright and wait for 60 seconds before repeating step 2 to 6.
8. Replace the mouthpiece cover by firmly pushing and snapping it into position.
9. Rinse your mouth with water after inhaling to minimise risk of oropharyngeal candida infections and the effect of bitter taste of the drug.

Important

- Do not rush step 4, 5, & 6.

- It is important that you start to breathe in as slowly as possible just before releasing the medication. Practise in front of a mirror for the first few times.
- If you see “mist” coming from the top of the inhaler or the sides of your mouth, you should start again from step 2. This escaping mist indicates failure of technique.
- If your doctor has given you different instruction for using your inhaler, please follow the instruction carefully.
- Inform or consult your doctor / pharmacist if you have any difficulties.

Parents must assist children who need help in using the inhaler correctly.

Cleaning your inhaler

Your inhaler should be cleaned at least once a week.

1. Remove the metal canister and mouthpiece cover from the actuator (plastic casing).
2. Rinse the actuator and mouthpiece cover thoroughly using warm water.
3. Dry the actuator and mouthpiece cover inside and out.
4. Replace the metal canister and mouthpiece cover.
5. Do not put the metal canister into water.

Disposal

Ask your doctor or pharmacist on how to dispose of used inhaler. Each inhaler contains 200 actuations and patients are advised to keep track on the total number of actuations taken. Always be sure to dispose of used inhaler responsibly, in the recommended way since some of the medicine may remain inside it. The used inhaler should not be disposed of via wastewater or household waste to ensure environmental safety.

Storage Conditions

Store below 30°C. Do not freeze. Protect from direct sunlight.

Keep out of reach of children

Dosage Forms and Packaging Available

Pressurised metered-dose inhaler. It contains 200 actuations per canister.

Shelf life:

The expiry date is indicated on the packaging.

Registration Number: MAL

Name and Address of Manufacturer and Product Registration Holder

Tenang Pharma Sdn Bhd
PT 3337, Jalan Permata 1/2, Arab Malaysian Industrial Park, 71800 Nilai, Negeri Sembilan Darul Khusus, MALAYSIA.

Revision Date: 23rd September 2022

Font size: 7 (as measured in Times New Roman)

BUDETEN is a registered trademark of Tenang
Pharma Sdn Bhd. All rights reserved.